COMPUTER SCIENCE TRIPOS Part IB, Part II 50% – 2022 – Paper 7

2 Economics, Law and Ethics (ah793)

You work for a consumer electronics company that dominates the market for mobile telephone handsets and operates an 'app store' for the distribution of mobile phone applications.

Information economics

(*a*) How does the market for mobile phone handsets differ from the market for mobile phone applications? [4 marks]

Answer: The main difference is the fixed and marginal costs (define for 2 marks). Software has a high fixed cost, but low marginal costs (the cost to produce one extra copy) (1 mark). Hardware manufacturers have capacity constraints, so additional factories, materials, and workers are required to produce hardware at scale (1 mark).

Information economics

(*b*) Why might a company providing hardware also start distributing applications? [6 marks]

Answer: Software can be a complementary asset that locks in customers (define for 1 mark). Customers may buy particular brands of hardware because of the applications that can run on it (1 mark). For example, they may buy an Apple watch so they can use applications from the Apple playstore. Switching costs then increase (define for 1 mark), as the customer not only has to buy the new piece of hardware, but the applications that run on it (1 mark). Furthermore, applications may also increase network effects (define for 1 mark), with people looking for interoperatability across devices and within their interpersonal networks (1 mark).

Information economics

(*c*) How might the company price the mobile phone handsets and applications? Provide arguments for and against the available options. [10 marks]

Answer: In order to sell to value, the company may choose to use price discrimination (1 mark). Pigou sets out three degrees of price discrimination, namely personalised pricing, versioning, and group pricing (define for 3 marks). The argument for price discrimination is efficiency. It provides access at a lower cost to those who might not otherwise be able to afford the product or service, while also allowing companies to mop up the available surplus (2 marks). Therefore, price discrimination is Pareto efficient, in that everyone is better off, but nobody is worse off (2 marks). However, price discrimination is also unpopular, as it drives the erosion of privacy, and people can be unhappy that they paid a higher price compared to others (2 marks). Some answers may also discuss how to monetise appstores. The current model is to demand 30% of app sales, as well as in-app payments for most apps, but this is being challenged in the courts.